

Elements of Statistics and Probability

Nabil Ahmed Uthso

School of Data and Computational Sciences

BRAC University

E-mail: nabil.ahmed@bracu.ac.bd

Table of Contents

1 Mathematical Expectation of Random Variables

2 Joint, Marginal and Conditional Probability Distribution

Mathematical Expectation

- Let X be a random variable with probability mass function $f(x)$ (if X is discrete), or density function $f(x)$ (if X is continuous). Let $g(x)$ be a function of the random variable X . Then, the mathematical expectation of the random variable $g(x)$ is defined by

$$E[g(x)] = \begin{cases} \sum g(x)f(x); & \text{if } X \text{ is discrete} \\ \int g(x)f(x)dx; & \text{if } X \text{ is continuous} \end{cases}$$

- $E[g(x)]$ is also known as the expected value of $g(x)$

Expectation of Discrete Random Variable

- Let X be a discrete random variable which can take a finite or infinite sequence of possible values $x_1, x_2, \dots$ with corresponding probabilities $f(x_1), f(x_2), \dots$; then the mathematical expectation of the random variable X , denoted by μ is defined as

$$\mu = E[X] = \sum_i x_i f(x_i)$$

Example

- Let X be a random variable with probability function defined as follows:

x_i	2	4	6	8
$f(x_i)$	2/10	1/10	4/10	3/10

- (a) Show that it is a valid pmf.
- (b) What is the expected value of X ?

Solution

(a) Valid pmf:

- $f(x) > 0 \forall x \in \{2, 4, 6, 8\}$
- $\sum_x f(x) = 2/10 + 1/10 + 4/10 + 3/10 = 1$

(b) Expectation:

$$\begin{aligned} E[X] &= \sum x_i f(x_i) \\ &= (2 \times 2/10) + (4 \times 1/10) + (6 \times 4/10) + (8 \times 3/10) \\ &= 5.6 \end{aligned}$$

Example 2

- Imagine a game in which, on any play, a player has a 20% chance of winning Tk. 30 and an 80% chance of losing Tk. 10. What is the expected gain/loss of the player in the long run?

Solution

- Let X = the gain on a play. The the pmf can be written as

x	30	-10
$f(x)$	0.2	0.8

$$\begin{aligned} E(X) &= \sum x_i f(x_i) \\ &= (30 \times 0.2) + (-10 \times 0.8) \\ &= -2 \end{aligned}$$

Example 3

- If the random variable X is the top face of a tossed, fair, six-sided die, what is the expected value of X ?

Solution

- The pmf of X is given by

$$f(x) = \frac{1}{6}; \text{ for } 1, 2, 3, 4, 5, 6$$

x	1	2	3	4	5	6
$f(x)$	1/6	1/6	1/6	1/6	1/6	1/6

$$\begin{aligned} E[X] &= \sum x_i f(x_i) \\ &= (1 \times 1/6) + (2 \times 1/6) + (3 \times 1/6) + (4 \times 1/6) + \\ &\quad (5 \times 1/6) + (6 \times 1/6) \\ &= 3.5 \end{aligned}$$

Expectation of Continuous Random Variables

- Let X be a continuous random variable with probability density function $f(x)$; then the mathematical expectation of the random variable X , denoted by μ is defined as

$$\mu = E[X] = \int_{-\infty}^{\infty} xf(x)$$

Example

- Suppose X is a continuous random variable with probability density function

$$f(x) = \begin{cases} 2x; & 0 \leq x \leq 1 \\ 0; & \text{otherwise} \end{cases}$$

(a) What is the expected value of X ?

Solution

- The expected value of X is given by

$$\begin{aligned} E[X] &= \int_0^1 xf(x)dx \\ &= \int_0^1 x \times 2x dx \\ &= \int_0^1 2x^2 dx \\ &= 2 \left[\frac{x^3}{3} \right]_0^1 \\ &= 2 \left(\frac{1}{3} - \frac{0}{3} \right) \\ &= \frac{2}{3} \end{aligned}$$

Exercise

- The pdf of a random variable X is defined as

$$f(x) = \begin{cases} x; & 0 \leq x < 1 \\ 2 - x; & 1 \leq x < 2 \\ 0; & x > 2 \end{cases}$$

- (a) What is the expected value of X ? (Ans: 1)

Properties of Mathematical Expectation

- Linearity properties:
 - $E[c] = c$, where c is a constant
 - $E[cX] = cE[X]$
 - $E[X + c] = E[X] + c$
 - $E[X + Y] = E[X] + E[Y]$
 - $E[X - Y] = E[X] - E[Y]$
- Multiplicity property:
 - If X and Y are two **independent** random variables, then

$$E[XY] = E[X] E[Y]$$

Variance of Random Variables

- Variance: $\text{Var}(X) = E(X^2) - [E(X)]^2$
- Standard deviation:

$$\text{SD}(X) = \sqrt{\text{Var}(X)} = \sqrt{E(X^2) - [E(X)]^2}$$

Properties of Variance

- $V[c] = 0$, where c is a constant
- $V[cX] = c^2 V[X]$
- $V[X + c] = V[X]$
- $V[X + Y] = V[X] + V[Y]$ if X and Y are independent
- $V[X - Y] = V[X] + V[Y]$ if X and Y are independent
- Coefficient of variation (CV): $CV(X) = \frac{SD(X)}{E(X)}$

Example

- You want to open a new Cafe. After your market research, you found that 20% of similar cafes make a monthly loss of Tk. 50,000, 30% of them make no profit or loss, 40% make a profit of Tk. 50,000, and 10% of them make a profit of Tk. 150,000.
 - (a) What is your expected profit if you decide to open a new Cafe?
 - (b) What is the standard deviation in your profit amount?
 - (c) What is the CV in your profit amount?
 - (d) If your fixed cost increases by Tk. 10,000, what will be your new expected profit?
 - (e) If your fixed cost increases by Tk. 10,000, what will be your new standard deviation of your profit?

Solution

- Let X be the profit amount. The pmf can be written as

x	-50,000	0	50,000	150,000
$f(x)$	0.2	0.3	0.4	0.1

(a)

$$\begin{aligned} E(X) &= (-50000 \times 0.2) + (0 \times 0.3) + (50000 \times 0.4) \\ &\quad + (150000 \times 0.1) \\ &= 25000 \end{aligned}$$

Solution

(b)

$$\begin{aligned} E(X^2) &= ((-50000)^2 \times 0.2) + (0^2 \times 0.3) + (50000^2 \times 0.4) \\ &\quad + (150000^2 \times 0.1) \\ &= 3750000000 \end{aligned}$$

- Therefore, $V(X) = E(X^2) - (E(X))^2 = 3125000000$
- Hence, $SD(X) = \sqrt{V(X)} = \sqrt{3125000000} = 55901.69$

Solution

(c)

$$\begin{aligned} CV &= \frac{SD(X)}{E(X)} \times 100\% \\ &= \frac{55901.69}{25000} \times 100\% \\ &= 22.36\% \end{aligned}$$

(d)

$$\begin{aligned} E(X - 10000) &= E(X) - E(10000) \\ &= 25000 - 10000 \\ &= 15000 \end{aligned}$$

Solution

(e)

$$\begin{aligned}V(X - 10000) &= V(X) - 0 \\&= V(X) \\&= 3125000000\end{aligned}$$

- Therefore,

$$SD(X - 10000) = \sqrt{V(X - 10000)} = \sqrt{3125000000} = 55901.69$$

Exercise

- Suppose X is a continuous random variable with probability density function

$$f(x) = \begin{cases} 2x; & 0 \leq x \leq 1 \\ 0; & \text{otherwise} \end{cases}$$

(a) Find $E(X)$, $V(X)$, $CV(X)$, $E(2 + 3X)$, $V(2 + 3X)$, $CV(2 + 3X)$

Table of Contents

- 1 Mathematical Expectation of Random Variables
- 2 Joint, Marginal and Conditional Probability Distribution

Joint Probability Distribution for Discrete R.V.

- Let X and Y be two discrete random variables and $P_{X,Y}(x, y)$ be a function of X and Y , then $P_{X,Y}(x, y)$ is called the joint probability mass function (pmf) if the following conditions are satisfied:

- $P_{X,Y}(x, y) \geq 0 \forall x, y$
- $\sum_x \sum_y P_{X,Y}(x, y) = 1$

Marginal Probability Distribution for Discrete R.V.

- If X and Y are two discrete random variables with joint pmf $P_{X,Y}(x, y)$ then
 - The marginal pmf of X is

$$P_X(x) = \sum_y P_{X,Y}(x, y)$$

- The marginal pmf of Y is

$$P_Y(y) = \sum_x P_{X,Y}(x, y)$$

Conditional Probability Dist. for Discrete R.V.

- If X and Y are two discrete random variables with joint pmf $P_{X,Y}(x,y)$ and marginal pmf $P_X(x)$ and $P_Y(y)$, then
 - The conditional probability distribution of X given Y can be written as

$$P_{X|Y}(x|y) = \frac{P_{X,Y}(x,y)}{P_Y(y)}$$

- The conditional probability distribution of Y given X can be written as

$$P_{Y|X}(y|x) = \frac{P_{X,Y}(x,y)}{P_X(x)}$$

Independence of Random Variables

- Jointly distributed random variables, say X and Y , are said to be independent if and only if their joint probability function is the product of their marginal probability functions. i.e.

$$P_{X,Y}(x,y) = P_X(x) P_Y(y)$$

Example

- The joint probability distribution of two random variables X and Y is as follows:

$X \setminus Y$	0	1	2
0	0.1	0	0.2
1	0.2	0.1	0.0
2	0.0	0.2	0.2

- Find the Marginal Probabilities of X and Y .
- Compute the expected values of X and Y .
- Determine if X and Y are independent.
- What is the probability that Y will be 2 given X is 1?

Solution

(a) Consider the following joint distribution:

$X \setminus Y$	0	1	2	$P_X(X = x)$
0	0.1	0	0.2	0.3
1	0.2	0.1	0.0	0.3
2	0.0	0.2	0.2	0.4
$P_Y(Y = y)$	0.3	0.3	0.4	1

- Therefore, the marginal distribution of X can be written as

x	0	1	2
$P(x)$	0.3	0.3	0.4

- Therefore, the marginal distribution of Y can be written as

y	0	1	2
$P(y)$	0.3	0.3	0.4

Solution

(b)

$$E(X) = \sum_x xP(x) = (0 \times 3) + (1 \times 0.3) + (2 \times 0.4) = 1.1$$

$$E(Y) = \sum_y yP(y) = (0 \times 3) + (1 \times 0.3) + (2 \times 0.4) = 1.1$$

Solution

(c) X and Y are independent if and only if $P_{X,Y}(x, y) = P(x) P(y)$.

We can show that $P_{X,Y}(x, y) \neq P(x) P(Y)$ for some X and Y

- Consider the situation when $X = 0$ and $Y = 0$, so

$$P_{X,Y}(x = 0, y = 0) = 0.1$$

- And $P(x = 0) \times P(y = 0) = 0.3 \times 0.3 = 0.09$
- Hence, $P_{X,Y}(x, y) \neq P(x) P(Y)$

(d)

$$P(y = 2|x = 1) = \frac{P(x = 1, y = 2)}{P(x = 1)} = \frac{0}{0.3} = 0$$

Exercise 1

- The joint probability distribution of Weather (W) and Temperature (T) is as follows

$W \setminus T$	Hot	Cold	$P(w)$
Sunny	3/10		1/2
Rainy		2/15	
Snowy			1/3
$P(T)$	1/3		

- Complete the probability distribution table.
- Are Weather and Temperature independent of each other?
- What is the probability that it will rain given it is cold?

Exercise 2

- The data collected on two random variables X and Y is as follows:

$X \setminus Y$	0	1	2	Total
0		0		
1	0	10		40
2	0			30
Total		25	55	100

- Complete the table.
- Find the joint probability distribution of X and Y .
- Find the Marginal distribution of X and Y .

Joint Probability Distribution for Continuous R.V.

- Let X and Y be two continuous random variables and $f_{X,Y}(x,y)$ be a function of X and Y , then $f_{X,Y}(x,y)$ is called the joint probability density function (pdf) if the following conditions are satisfied:
 - $f_{X,Y}(x,y) \geq 0 \forall x,y$
 - $\int_x \int_y f_{X,Y}(x,y) dy dx = 1$

Marginal Probability Dist. for Continuous R.V.

- If X and Y are two continuous random variables with joint pdf $f_{X,Y}(x, y)$ then

- The marginal pdf of X is

$$f_X(x) = \int_y f_{X,Y}(x, y) dy$$

- The marginal pdf of Y is

$$f_Y(y) = \int_x f_{X,Y}(x, y) dx$$

Conditional Probability Dist. for Discrete R.V.

- If X and Y are two continuous random variables with joint pdf $f_{X,Y}(x,y)$ and marginal pdf $f_X(x)$ and $f_Y(y)$, then
 - $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$
 - $f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$

Example

- Consider the joint pdf of the random variables X and Y :

$$f(x, y) = 4xy; 0 \leq x \leq 1, 0 \leq y \leq 1$$

- (a) Show that it is a pdf.
- (b) Find the probability that X is at most $1/2$ and Y is not more than $3/4$.
- (c) Find the marginal distribution of X .
- (d) Obtain $E(X)$, $V(X)$, and $CV(X)$.
- (e) Find the conditional probability that Y is bigger than $2/3$ given X is less than $2/3$.

Thank You